

LAB 3 & 4 NOTES

There are examples below that talk about floating point numbers, IEEE754 format and their conversion. The examples explain the steps involved.

1. Convert the decimal number 23.75 to its IEEE 754 single-precision representation.

Determine the sign bit (bit 31): Since 23.75 is positive, the sign bit is 0.

Convert the integer part (23) to binary: 23 in binary is 10111.

Convert the fractional part (0.75) to binary: 0.75 in binary is 0.11.

Normalize the binary representation: Shift the binary point left, and adjust the exponent accordingly.

101.11 (normalized) = 1.0111×2^4

Calculate the exponent value: Exponent = 4. Add the bias for single-precision (127) to the exponent: $4 + 127 = 131$ (in binary, 10000011).

Combine the sign bit, exponent, and mantissa to get the final IEEE 754 representation:

Sign bit (0) | Exponent (10000011) | Mantissa (011100000000000000000000)

So, the IEEE 754 single-precision representation of 23.75 is:
0 10000011 011100000000000000000000

2. Convert the IEEE 754 binary representation 010000001011100000000000000000 to its decimal equivalent.

Sign bit (bit 31) is 0, indicating a positive number.

Exponent bits (bits 30-23) are 10000011, which is 131 in decimal.

Mantissa bits (bits 22-0) are 011100000000000000000000 in binary.

To convert the mantissa to decimal, we consider it as a fraction:

Mantissa (binary) = $0.011100000000000000000000$

Now, apply the formula for IEEE 754 conversion:

$$\text{Decimal} = (-1)^{\text{sign}} * 2^{(\text{exponent} - \text{bias})} * 1.\text{mantissa}$$

Where:

The sign is 0 (positive).

exponent is 131.

bias for single-precision is 127.

$$\text{Decimal} = 1 * 2^{(131 - 127)} * 1.011100000000000000000000 \text{ (as a fraction)}$$

$$\text{Decimal} = 2^4 * 1.4375$$

$$\text{Decimal} = 16 * 1.4375$$

$$\text{Decimal} = 22.8$$

So, the decimal equivalent of the given IEEE 754 binary representation is 22.8.

3. Convert 61.3471 into binary.

Convert the integer part (61) to binary:

61 in binary is 111101.

Convert the fractional part (0.3471) to binary:

To convert the fractional part to binary, you can use the following steps:

Multiply the fractional part by 2.

The whole number part of the result becomes the next binary digit.

Keep repeating the process, multiplying the remaining fractional part by 2 and noting the whole number part until you have the desired number of binary digits.

For 0.3471:

$$0.3471 * 2 = 0.6942, \text{ so the first binary digit is 0.}$$

$$0.6942 * 2 = 1.3884, \text{ so the second binary digit is 1.}$$

$$0.3884 * 2 = 0.7768, \text{ so the third binary digit is 0.}$$

$0.7768 * 2 = 1.5536$, so the fourth binary digit is 1.

$0.5536 * 2 = 1.1072$, so the fifth binary digit is 1.

So, the fractional part in binary is 0.01111.

Combine the binary representations of the integer and fractional parts:

61.3471 in binary is 111101.01111.

So, the binary representation of 61.3471 is 111101.01111.

4. Convert 111101.01111 into decimal.

Separate the number into its integer and fractional parts:

Integer part: 111101

Fractional part: 01111

Convert the integer part to decimal using the binary to decimal conversion method:

Integer part in binary: 111101

Decimal value of the integer part:

$$\begin{aligned} &1 * 2^5 + 1 * 2^4 + 1 * 2^3 + 1 * 2^2 + 0 * 2^1 + 1 * 2^0 \\ &= 32 + 16 + 8 + 4 + 0 + 1 \\ &= 61 \end{aligned}$$

Convert the fractional part to decimal using the binary to decimal conversion method:

Fractional part in binary: 01111

Decimal value of the fractional part:

$$\begin{aligned} &0 * 2^{(-1)} + 1 * 2^{(-2)} + 1 * 2^{(-3)} + 1 * 2^{(-4)} + 1 * 2^{(-5)} \\ &= 0 + 0.25 + 0.125 + 0.0625 + 0.03125 \\ &= 0.46875 \end{aligned}$$

Combine the decimal values of the integer and fractional parts:

Decimal value of 111101.01111 = Integer part (61) + Fractional part (0.46875)

Decimal value = $61 + 0.46875 = 61.46875$

So, the decimal equivalent of the binary number 111101.01111 is 61.46875.